MiniMax-H3: From 653.838 Seconds to 14.561 Seconds

A cumulative, evidence-backed latency optimization report for 15.083-second text-to-video-with-audio generation

NVIDIA RTX PRO 5000-class SM120 x8 | vLLM-Omni | FastH3 distillation | VSA | FP8 | TAEH3 | valid H.264/AAC MP4

FINAL FORMAL RESULT

Mean E2E	Samples	Speedup	Reduction	Target margin	RTF
14.561 s	14.576 / 14.555 / 14.551 s	44.903x	97.773%	0.439 s	0.965

Generated 2026-09-07 19:02 UTC. Benchmark optimization stopped after D6; D7 remains unmeasured.

Claim policy: measured, inferred, implemented-only, and rejected results are visually distinguished. No missing profile is reconstructed or simulated.

Executive summary

The final three-run mean is 14.561 s for a 15.083333 s deliverable: faster than real time and 44.903x faster than the 653.838 s original-H3 control.

Most of the speedup came from two deliberate approximation boundaries: four-forward FastH3 distillation and visual sparse attention. Exact systems work then removed layout copies, communication setup, device-to-host volume, and serialization. Calibrated FP8/E4M3 reduced DiT cost. TAEH3 removed the full patch-parallel VAE from the critical path, and cross-rank audio overlap supplied the final margin.

Question	Answer
Did the benchmark satisfy the exact endpoint contract?	Yes: POST through a validated 1280x704, 24 fps, 362-frame H.264 video with stereo 32 kHz AAC.
Was the final result a single best sample?	No: 14.561 s is the arithmetic mean of 14.576, 14.555, and 14.551 s.
Are all changes lossless?	No. Distillation, VSA, FP8/E4M3, and TAEH3 are explicitly marked as lossy/approximate.
Was a final decoder Nsight trace captured?	No. NVTX instrumentation exists, but the D7 profiling attempt was stopped before model load after GPUs were occupied e
Can exact optimizations be added again?	No. The final D6 already contains the retained exact-general VSA and systems changes.

Claims, labels, and evidence discipline

This report uses four evidence labels so that cumulative numbers cannot silently become stronger claims than the artifacts support.

Label	Meaning	Allowed conclusion
Measured	A persisted benchmark, profiler, media, or metric artifact exists.	Quote the value with its stated scope.
Exact	Operation is intended to preserve the selected pipeline's numerical/media boundary.	Attribute latency, not perceptual improvement.
Lossy	Sampling graph, numeric precision, or decoder changed.	Require screenshots and quantitative diagnostics.
Implemented-only	Code/instrumentation exists without a valid formal timing.	No latency claim.
Rejected	Matched result was slower, unstable, contaminated, or crossed an approved quality boundary.	Use as negative evidence, never add to the cumulative path.

The Nsight category tables sum GPU kernel durations across ranks and streams. They are composition views, not wall-clock decompositions. Barrier residency is synchronization/spin residency, not a direct measurement of bytes-on-wire.

Fixed benchmark contract

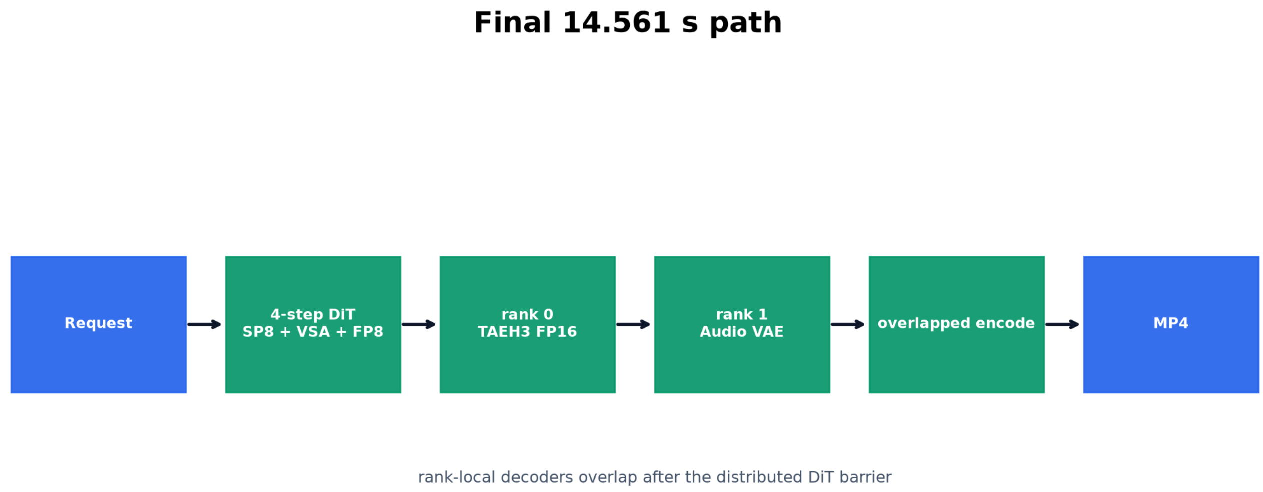
Field	Locked value
Hardware	8x NVIDIA RTX PRO 5000-class SM120 GPU; node-default/unlocked clocks
Distributed shape	DiT TP1/SP8; text encoder Qwen TP8; original baseline retained its documented control topology
Request	Text-to-video-with-audio, seed 1101, 1280x720 request
Decoded output	1280x704, 24 fps, 362 frames, 15.083333 s
Audio	Stereo, 32 kHz AAC
FastH3 schedule	4 forwards, sigma [0.999, 0.749, 0.5, 0.25, 0]
E2E boundary	HTTP POST start through a probe-valid H.264/AAC MP4
Acceptance statistic	Three repeated formal samples where available; arithmetic mean
Target	Strictly less than 15.000 seconds

Output height 704 is the decoder-valid realization of the 720-pixel request. The media contract, not the request field alone, is what was verified.

Timing semantics: what 14.561 seconds includes

The timer includes request handling, distributed DiT work, video decoding, audio decoding, device-to-host transfers, H.264/AAC encoding and MP4 muxing. It stops only after the returned file passes the fixed media probe.

It does not include model cold start, weights download, report generation, or offline quality analysis. Profiler overhead is excluded from D6; profiling runs are used only for bottleneck localization.



The final online request path and endpoint boundary.

Official MiniMax-H3 prompt-writing guide

The guide was read from MiniMax-AI/MiniMax-H3 at commit `d21241f0a4b3acbb34c97dae47fa417b7065e438`. Its T2VA structure is: integrated multimodal description, overall soundscape, then non-diegetic music. It asks for concrete shot composition, subject, action, environment, camera motion type/amplitude/speed, timed later shots, and a total duration of 4-15 seconds.

```
Source: https://github.com/MiniMax-AI/MiniMax-H3/tree/d21241f0a4b3acbb34c97dae47fa417b7065e438/skills/h
3-prompt-writing
```

The calibration prompt predates that guide and was hash-locked. It was not rewritten for benchmark execution. The guide is therefore used as a documentation lens, not as evidence that a different prompt produced the reported timing.

Executed prompt and guide-conformant semantic rendering

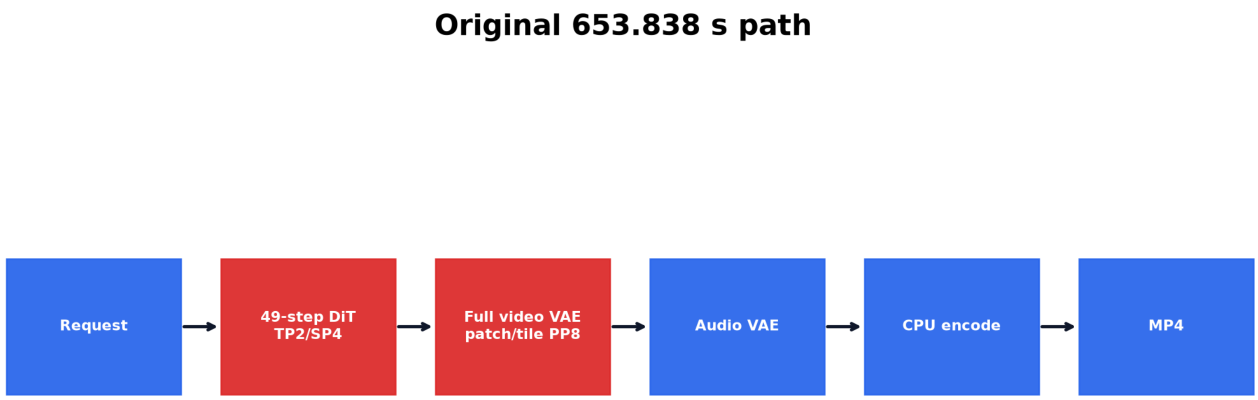
Executed UTF-8 prompt SHA-256:
`613e0a1beccf83566e3531852c52c2e11daffa3b255222cd67726cdf73c1a563`

Faithful English rendering of the locked prompt: A live-action handheld smartphone shot in a small kitchen at dusk blends with luminous hand-drawn 2D animation. Sunset light remains at the window. An old wooden table, a half-washed mug, a fogged glass bottle, and a hanging rag create an ordinary lived-in space. One-handed shake, hesitant close focus, and backlight exposure pumping make the event feel caught in a rush rather than polished advertising. Audio contains only kitchen ambience, soft electronic sounds from the illustrated creature, and its tiny calls.

Guide-field documentation only (not executed input):

Guide field	Documentation rendering
integrated_multimodal_description	Single continuous 15 s handheld close shot; live-action kitchen, dusk backlight, glowing 2D creature, hesitant rack focus a
overall_soundscape	Diegetic kitchen room tone, dish and cloth detail, soft electronic creature tones and tiny calls.
non_diegetic_music	None.
camera	Smartphone, one-handed micro-shake, close-focus hesitation, natural exposure fluctuation; no commercial polish.

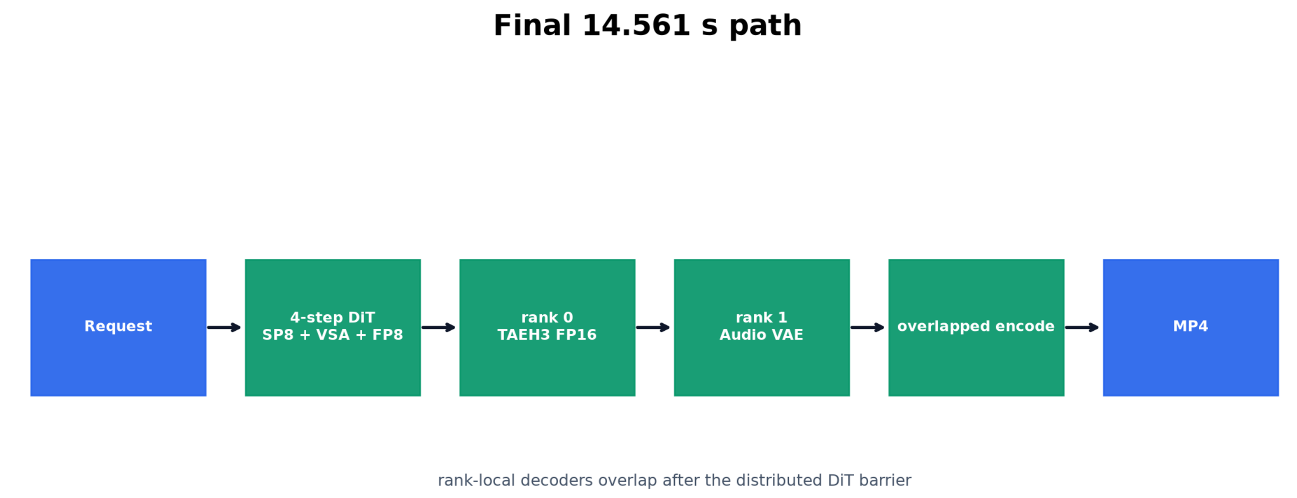
Original architecture: where 653.838 seconds went



The original-H3 control used 49 diffusion forwards and a collective-heavy full video VAE.

DiT alone consumed 633.058 s. The first accepted optimizations attacked distributed overhead without changing the original model. Nsight later showed cuDNN attention as 64.53% of summed GPU kernel time in the profiled step, followed by GEMM at 19.82% and RDMA-barrier residency at 14.36%.

Final architecture: why 14.561 seconds is feasible

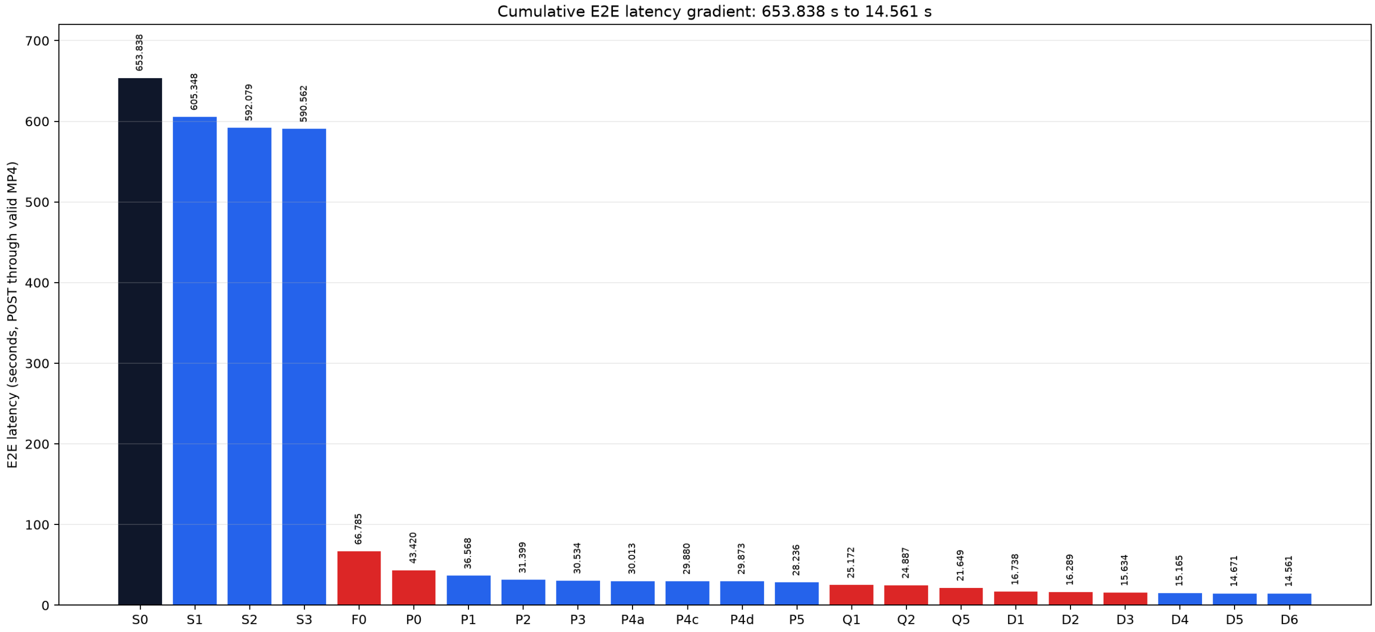


Four DiT forwards, sparse attention, low-precision GEMMs and rank-local parallel decoders form the final path.

The final design is not one magic kernel. It is a critical-path rewrite: remove 45 forwards, execute a smaller attention graph, reduce bytes and layouts, replace the full video VAE, and overlap independent terminal work on separate ranks.

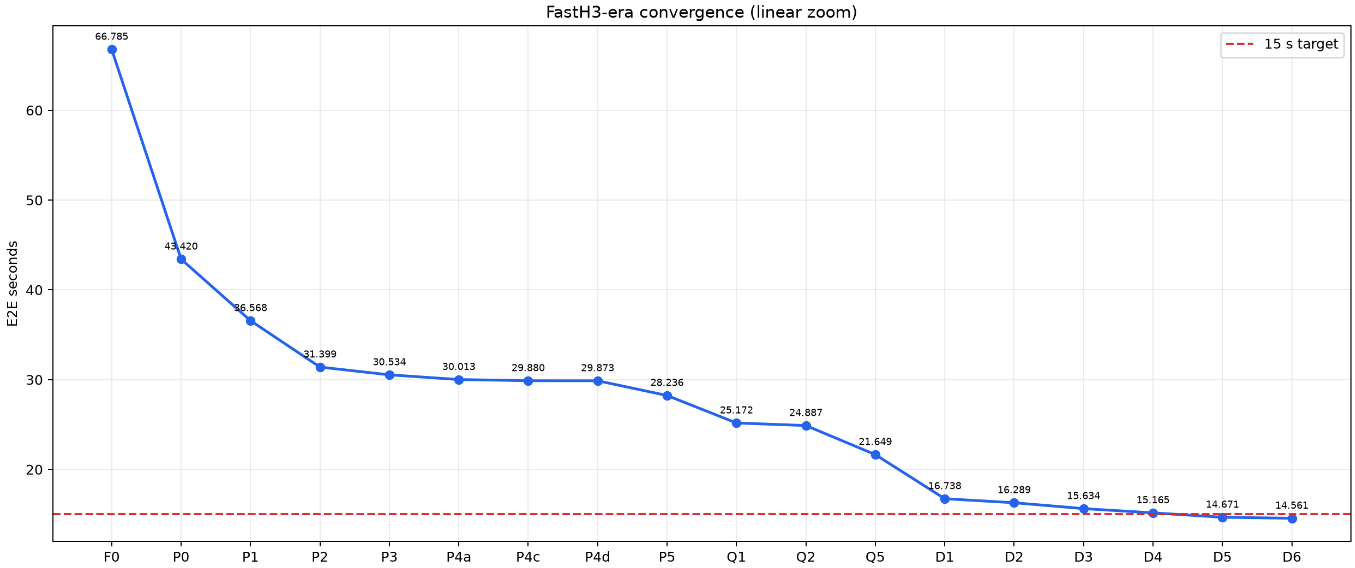
Standalone E2E gradient bar chart

Requested standalone visualization: every accepted cumulative stage is a bar, descending from 653.838 s to 14.561 s.



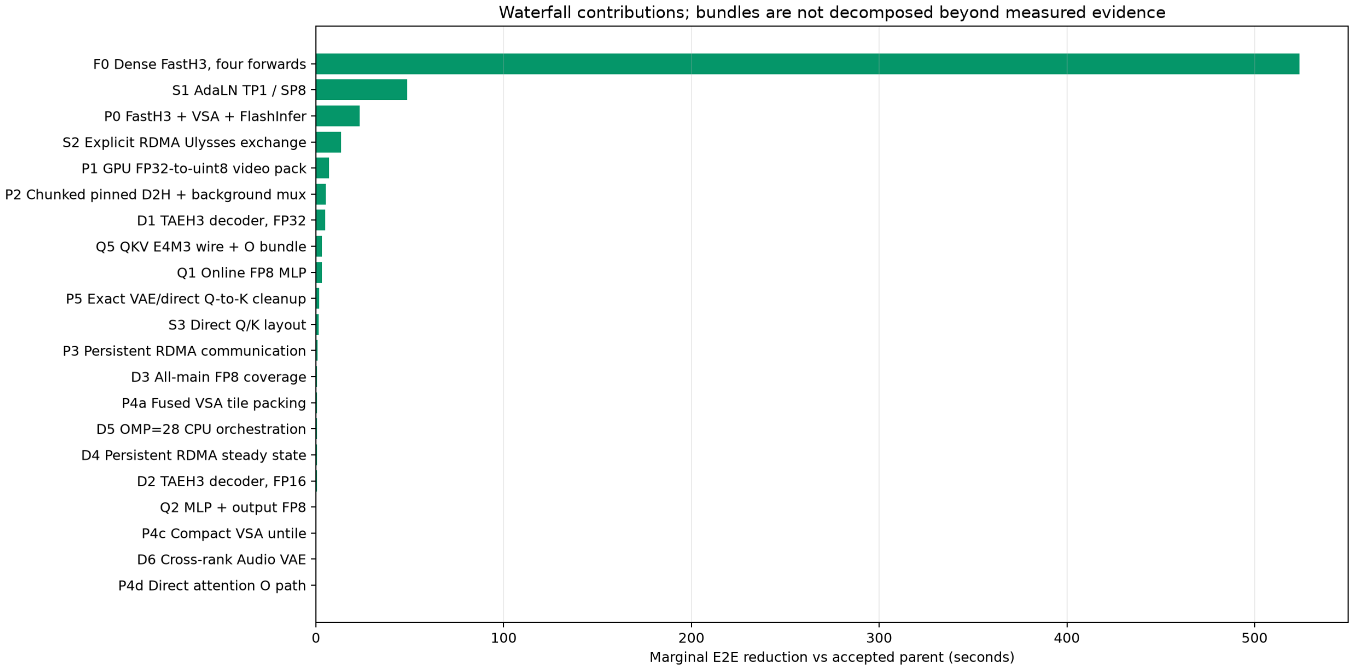
Red bars mark lossy boundary entries; blue bars are exact/system changes inside the selected pipeline.

FastH3-era convergence



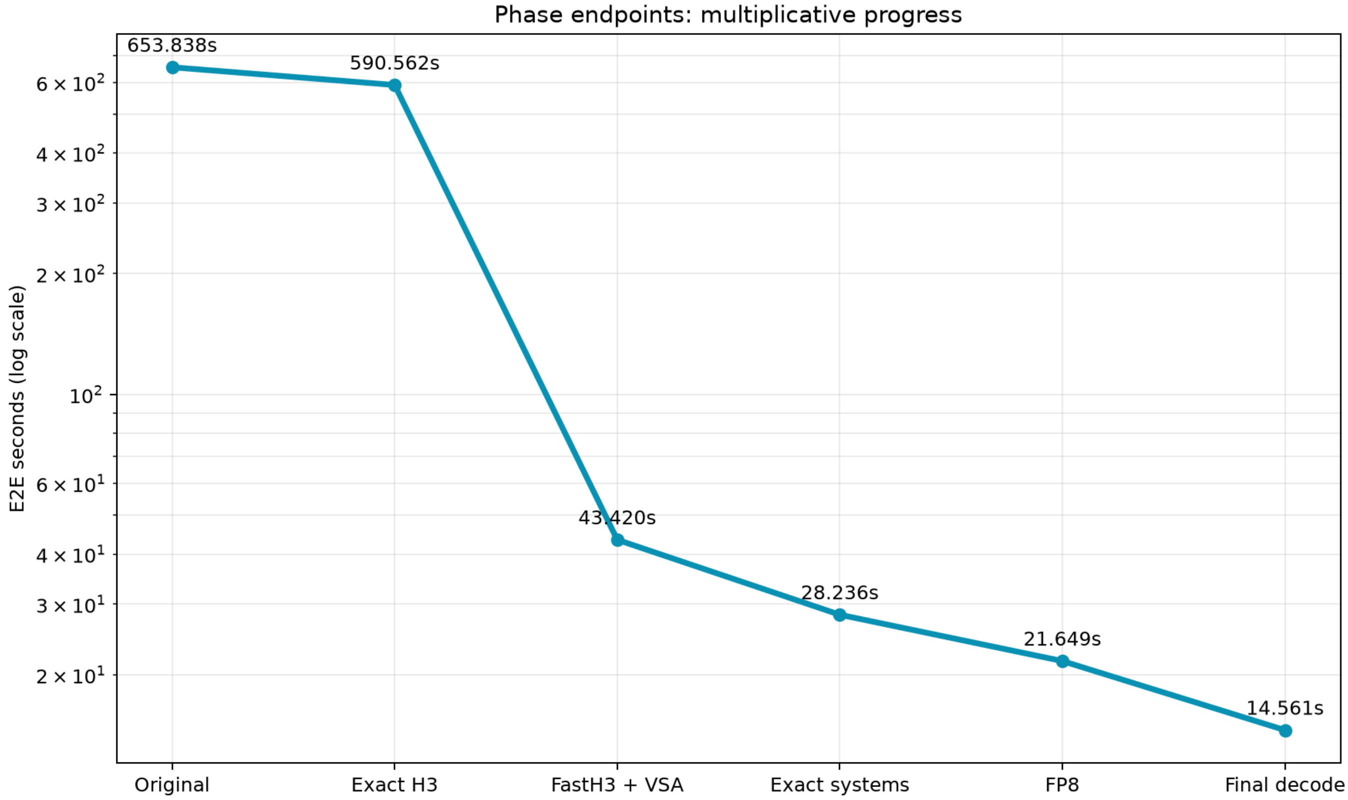
Linear zoom after distillation; the 15-second target is shown independently of the 653-second baseline.

Marginal contribution waterfall



Each bar is the observed difference from the accepted parent. Mixed bundles are not falsely decomposed.

Optimization phases on a logarithmic scale



Phase endpoints expose the multiplicative nature of the speedup.

FastH3 distillation dominates absolute seconds. Once below 70 seconds, decoder and systems work become first-order. Near 15 seconds, even a 0.1-second overlap matters, so repeated samples and clean profiler-off measurements become essential.

Accepted cumulative path

ID	Optimization	E2E s	Delta s	Speedup vs S0	Fidelity
S0	Original H3 control	653.838	-	1.00x	Original
S1	AdaLN TP1 / SP8	605.348	48.490	1.08x	Exact
S2	Explicit RDMA Ulysses exchange	592.079	13.269	1.10x	Exact
S3	Direct Q/K layout	590.562	1.517	1.11x	Exact
F0	Dense FastH3, four forwards	66.785	523.777	9.79x	Lossy
P0	FastH3 + VSA + FlashInfer	43.420	23.365	15.06x	Lossy
P1	GPU FP32-to-uint8 video pack	36.568	6.852	17.88x	Exact
P2	Chunked pinned D2H + background mux	31.399	5.169	20.82x	Exact
P3	Persistent RDMA communication	30.534	0.865	21.41x	Exact
P4a	Fused VSA tile packing	30.013	0.521	21.79x	Exact
P4c	Compact VSA untile	29.880	0.133	21.88x	Exact
P4d	Direct attention O path	29.873	0.007	21.89x	Exact / noise-sized
P5	Exact VAE/direct Q-to-K cleanup	28.236	1.637	23.16x	Exact
Q1	Online FP8 MLP	25.172	3.064	25.97x	Lossy
Q2	MLP + output FP8	24.887	0.285	26.27x	Lossy
Q5	QKV E4M3 wire + O bundle	21.649	3.238	30.20x	Lossy
D1	TAEH3 decoder, FP32	16.738	4.911	39.06x	Lossy
D2	TAEH3 decoder, FP16	16.289	0.449	40.14x	Lossy
D3	All-main FP8 coverage	15.634	0.655	41.82x	Lossy
D4	Persistent RDMA steady state	15.165	0.469	43.11x	Exact
D5	OMP=28 CPU orchestration	14.671	0.494	44.57x	Exact
D6	Cross-rank Audio VAE	14.561	0.110	44.90x	Exact

Only this parent chain appears in the standalone gradient chart. Side branches are preserved separately and never stacked.

Optimization 01 — S1: AdaLN TP1 / SP8

653.838 s -> 605.348 s | measured reduction 48.490 s | cumulative speedup 1.080x

Dimension	Evidence-backed statement
Phase	A
Status	accepted
Fidelity	Exact
Principle	Replicate small AdaLN tensors and spend all eight ranks on sequence parallelism.
Evidence	DiT 584.707 s; -48.490 s E2E.
Interpretation	Removes tensor-parallel collectives from a latency-dominated batch-one request.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 02 — S2: Explicit RDMA Ulysses exchange

605.348 s -> 592.079 s | measured reduction 13.269 s | cumulative speedup 1.104x

Dimension	Evidence-backed statement
Phase	A
Status	accepted
Fidelity	Exact
Principle	Move the large sequence exchange through the topology-aware RDMA path.
Evidence	DiT 571.264 s; -13.269 s E2E.
Interpretation	The gain is interconnect-path selection, not model approximation.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 03 — S3: Direct Q/K layout

592.079 s -> 590.562 s | measured reduction 1.517 s | cumulative speedup 1.107x

Dimension	Evidence-backed statement
Phase	A
Status	accepted
Fidelity	Exact
Principle	Produce the attention layout directly and delete redundant transposes/copies.
Evidence	DiT 569.830 s; -1.517 s E2E.
Interpretation	A small but repeatable layout win before distillation.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 04 — F0: Dense FastH3, four forwards

590.562 s -> 66.785 s | measured reduction 523.777 s | cumulative speedup 9.790x

Dimension	Evidence-backed statement
Phase	B
Status	accepted
Fidelity	Lossy
Principle	Use the distilled four-step sigma schedule [0.999, 0.749, 0.5, 0.25, 0].
Evidence	DiT 46.988 s; -523.777 s E2E.
Interpretation	This is the largest single change and the first explicit quality boundary.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This optimization crosses a declared approximation boundary. The dedicated quality section contains screenshots and full-video post-codec metrics for the nearest available matched artifact pair.

Optimization 05 — P0: FastH3 + VSA + FlashInfer

66.785 s -> 43.420 s | measured reduction 23.365 s | cumulative speedup 15.058x

Dimension	Evidence-backed statement
Phase	B
Status	accepted
Fidelity	Lossy
Principle	Apply visual sparse attention with exact FlashInfer sparse-kernel execution inside the distilled trajectory.
Evidence	DiT 22.704 s; -23.365 s E2E.
Interpretation	VSA changes the attention graph; the kernel itself is exact for that sparse graph.

Causality boundary

This rung contains a measured bundle. The E2E delta is attributed to the bundle only; component-level claims come from separate microbenchmarks or profiler evidence and are not summed into a fictional decomposition.

Quality handling

This optimization crosses a declared approximation boundary. The dedicated quality section contains screenshots and full-video post-codec metrics for the nearest available matched artifact pair.

Optimization 06 — P1: GPU FP32-to-uint8 video pack

43.420 s -> 36.568 s | measured reduction 6.852 s | cumulative speedup 17.880x

Dimension	Evidence-backed statement
Phase	C
Status	accepted
Fidelity	Exact
Principle	Quantize display pixels on GPU and copy uint8 instead of float frames.
Evidence	-6.852 s E2E.
Interpretation	Semantics are unchanged at the MP4 contract boundary.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 07 — P2: Chunked pinned D2H + background mux

36.568 s -> 31.399 s | measured reduction 5.169 s | cumulative speedup 20.824x

Dimension	Evidence-backed statement
Phase	C
Status	accepted
Fidelity	Exact
Principle	Overlap pinned host copies and CPU PyAV encode/mux with remaining device work.
Evidence	-5.169 s E2E.
Interpretation	The POST endpoint still waits for a valid H.264/AAC MP4.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 08 — P3: Persistent RDMA communication

31.399 s -> 30.534 s | measured reduction 0.865 s | cumulative speedup 21.413x

Dimension	Evidence-backed statement
Phase	C
Status	accepted
Fidelity	Exact
Principle	Reuse communication resources across layer iterations instead of reinitializing them.
Evidence	-0.865 s E2E.
Interpretation	Reduces orchestration and allocator overhead.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 09 — P4a: Fused VSA tile packing

30.534 s -> 30.013 s | measured reduction 0.521 s | cumulative speedup 21.785x

Dimension	Evidence-backed statement
Phase	C
Status	accepted
Fidelity	Exact
Principle	Fuse tile selection and packing while preserving selected values and order.
Evidence	Approximately -0.490 s in matched trials.
Interpretation	A kernel-launch and memory-traffic optimization.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 10 — P4c: Compact VSA untile

30.013 s -> 29.880 s | measured reduction 0.133 s | cumulative speedup 21.882x

Dimension	Evidence-backed statement
Phase	C
Status	accepted
Fidelity	Exact
Principle	Write only the compact valid output domain during sparse-attention reconstruction.
Evidence	-0.133 s E2E.
Interpretation	Avoids materializing padding that no downstream operator consumes.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 11 — P4d: Direct attention O path

29.880 s -> 29.873 s | measured reduction 0.007 s | cumulative speedup 21.887x

Dimension	Evidence-backed statement
Phase	C
Status	accepted
Fidelity	Exact / noise-sized
Principle	Route attention output directly into its consumer.
Evidence	-0.007 s E2E.
Interpretation	Retained for structural simplicity; the measured delta is within noise.

Causality boundary

The measured delta is noise-sized. The change is retained for simpler dataflow or because later stages depend on it, not because this single E2E pair proves a stable speedup.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 12 — P5: Exact VAE/direct Q-to-K cleanup

29.873 s -> 28.236 s | measured reduction 1.637 s | cumulative speedup 23.156x

Dimension	Evidence-backed statement
Phase	C
Status	accepted
Fidelity	Exact
Principle	Consolidate exact decoder and Q-to-K dataflow cleanups.
Evidence	Q0 control: DiT 21.347 s.
Interpretation	A mixed cleanup bundle; it is not used to assign per-kernel causality.

Causality boundary

This rung contains a measured bundle. The E2E delta is attributed to the bundle only; component-level claims come from separate microbenchmarks or profiler evidence and are not summed into a fictional decomposition.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 13 — Q1: Online FP8 MLP

28.236 s -> 25.172 s | measured reduction 3.064 s | cumulative speedup 25.975x

Dimension	Evidence-backed statement
Phase	D
Status	accepted
Fidelity	Lossy
Principle	Use online FP8 for MLP GEMMs while retaining wider precision where sensitivity is higher.
Evidence	DiT 18.522 s; -3.064 s E2E.
Interpretation	First low-precision compute boundary.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This optimization crosses a declared approximation boundary. The dedicated quality section contains screenshots and full-video post-codec metrics for the nearest available matched artifact pair.

Optimization 14 — Q2: MLP + output FP8

25.172 s -> 24.887 s | measured reduction 0.285 s | cumulative speedup 26.272x

Dimension	Evidence-backed statement
Phase	D
Status	accepted
Fidelity	Lossy
Principle	Extend FP8 to output projections with calibrated scaling.
Evidence	DiT 18.124 s; -0.285 s E2E.
Interpretation	A conservative expansion before QKV/wire bundling.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This optimization crosses a declared approximation boundary. The dedicated quality section contains screenshots and full-video post-codec metrics for the nearest available matched artifact pair.

Optimization 15 — Q5: QKV E4M3 wire + O bundle

24.887 s -> 21.649 s | measured reduction 3.238 s | cumulative speedup 30.202x

Dimension	Evidence-backed statement
Phase	D
Status	accepted
Fidelity	Lossy
Principle	Transmit QKV in E4M3 and bundle output communication to reduce bytes and launch count.
Evidence	DiT 14.994 s; -3.238 s E2E.
Interpretation	The accepted precision/communication endpoint before decoder replacement.

Causality boundary

This rung contains a measured bundle. The E2E delta is attributed to the bundle only; component-level claims come from separate microbenchmarks or profiler evidence and are not summed into a fictional decomposition.

Quality handling

This optimization crosses a declared approximation boundary. The dedicated quality section contains screenshots and full-video post-codec metrics for the nearest available matched artifact pair.

Optimization 16 — D1: TAEH3 decoder, FP32

21.649 s -> 16.738 s | measured reduction 4.911 s | cumulative speedup 39.063x

Dimension	Evidence-backed statement
Phase	E
Status	accepted
Fidelity	Lossy
Principle	Replace the full patch-parallel video VAE with the rank-local temporal autoencoder decoder.
Evidence	Decode 1.554 s; -4.911 s E2E.
Interpretation	Removes decoder collectives and most video decode compute.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This optimization crosses a declared approximation boundary. The dedicated quality section contains screenshots and full-video post-codec metrics for the nearest available matched artifact pair.

Optimization 17 — D2: TAEH3 decoder, FP16

16.738 s -> 16.289 s | measured reduction 0.449 s | cumulative speedup 40.140x

Dimension	Evidence-backed statement
Phase	E
Status	accepted
Fidelity	Lossy
Principle	Run TAEH3 convolution weights and activations in FP16.
Evidence	Mean E2E 16.289 s; decode 1.050 s.
Interpretation	Post-codec FP32-vs-FP16 quality is separately audited.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This optimization crosses a declared approximation boundary. The dedicated quality section contains screenshots and full-video post-codec metrics for the nearest available matched artifact pair.

Optimization 18 — D3: All-main FP8 coverage

16.289 s -> 15.634 s | measured reduction 0.655 s | cumulative speedup 41.822x

Dimension	Evidence-backed statement
Phase	E
Status	accepted
Fidelity	Lossy
Principle	Extend calibrated FP8 coverage across the remaining main DiT GEMM path.
Evidence	DiT 14.443 s; decode 1.003 s.
Interpretation	Completes the selected low-precision compute policy.

Causality boundary

This rung contains a measured bundle. The E2E delta is attributed to the bundle only; component-level claims come from separate microbenchmarks or profiler evidence and are not summed into a fictional decomposition.

Quality handling

This optimization crosses a declared approximation boundary. The dedicated quality section contains screenshots and full-video post-codec metrics for the nearest available matched artifact pair.

Optimization 19 — D4: Persistent RDMA steady state

15.634 s -> 15.165 s | measured reduction 0.469 s | cumulative speedup 43.115x

Dimension	Evidence-backed statement
Phase	E
Status	accepted
Fidelity	Exact
Principle	Keep the optimized communication path persistent through the formal repeated run.
Evidence	15.146/15.178/15.170 s; mean 15.165 s.
Interpretation	First result close to the 15-second target.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 20 — D5: OMP=28 CPU orchestration

15.165 s -> 14.671 s | measured reduction 0.494 s | cumulative speedup 44.567x

Dimension	Evidence-backed statement
Phase	E
Status	accepted
Fidelity	Exact
Principle	Right-size host parallelism for decode/encode orchestration on this node.
Evidence	14.680/14.672/14.660 s; mean 14.671 s.
Interpretation	First formal three-run mean below 15 seconds.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 21 — D6: Cross-rank Audio VAE

14.671 s -> 14.561 s | measured reduction 0.110 s | cumulative speedup 44.903x

Dimension	Evidence-backed statement
Phase	E
Status	accepted
Fidelity	Exact
Principle	Decode video with TAEH3 on rank 0 while rank 1 independently runs Audio VAE.
Evidence	14.576/14.555/14.551 s; mean 14.561 s.
Interpretation	Final accepted result; profiler disabled, valid MP4 returned.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Side branches: original-H3 and early FastH3

Family	Experiment	E2E s	Disposition
Exact-H3 side control	TP2/SP4 communication control	646.158	Not on accepted path
Exact-H3 side control	Q chunk = 2	587.398	Faster single side result; not retained as cumulative parent
FastH3 reference	Reference VSA	47.527	Diagnostic reference
FastH3 reference	FlashInfer-only contaminated E2E	52.965	DiT 27.108 s remains useful; E2E rejected

The 52.965-second FlashInfer-only E2E run was contaminated, so it is not an accepted endpoint. Its DiT measurement (27.108 s) remains a useful localization datum. The qchunk=2 exact-H3 side result was not inherited as the parent of later cumulative stages.

Side branches: FP8 and communication combinations

Family	Experiment	E2E s	Disposition
FP8 branch	MLP + QKV E4M3	23.930	Branch, not additive with Q2
FP8 branch	MLP + output + wire	23.679	Three-run mean branch
FP8 branch	+ O bundle	23.168	Three-run mean branch
FP8 branch	MLP + output + QKV compute	23.260	Alternate branch
FP8 branch	+ BF16 wire	22.333	Alternate branch

Q3a/Q4a and Q3b/Q4b/Q5 are alternative precision/communication branches. Treating their deltas as additive would overstate the speedup. Q5 is the only branch endpoint inherited by D1.

Rejected or non-retained experiments — set 1

Experiment	Measured result	Why it was not retained
Layerwise offload	~50 s per diffusion step	PCIe transfers dominated; categorically incompatible with the latency target.
LSA / SymmetricMemory	Unstable or slower	The topology/runtime combination did not produce a reliable improvement.
Hybrid attention path	30.625 vs 30.075 s	Added dispatch and layout cost.
Q chunk variant	32.240 vs 31.965 s	More chunk overhead than locality benefit.
Single-output path	43.575 vs 43.420 s	No measurable win.
NVENC	49.817 vs PyAV 49.494 s	Slower and produced a different encoded artifact.

Negative results are part of the optimization map: they prevent rediscovery and show that smaller tensors, more overlap, more fusion, or a different encoder are not automatically faster at this shape.

Rejected or non-retained experiments — set 2

Experiment	Measured result	Why it was not retained
VAE temporal prune	30.518 vs 30.503 s	No meaningful speedup.
Alternative GEMM backends	within +/-1.5%	No robust end-to-end advantage.
FC1 + SwiGLU CuTeDSL	4.72% slower	Fusion did not compensate for the generated kernel's efficiency loss.
Gate overlap	21.714 vs 21.676 s	Noise-sized regression.
Head-major VSA	15.785 s	Layout change regressed the final stack.
KV64	15.320 s	Reduced work did not offset scheduling/conversion overhead.

Negative results are part of the optimization map: they prevent rediscovery and show that smaller tensors, more overlap, more fusion, or a different encoder are not automatically faster at this shape.

Rejected or non-retained experiments — set 3

Experiment	Measured result	Why it was not retained
KV64 without LSE	15.217 s mean	Still slower than the exact-general selected path.
Same-GPU audio overlap	15.233 s	Video and audio competed for one GPU.
Adaptive partial-N32	+0.645%	Extra branch/scheduling cost.
KV64 to KV32 / KV16	+8.3% / +41.7%	Occupancy and useful-work balance collapsed.
Skip softmax	Approximate and slower	A lossy change with no latency benefit.
VAE batch cap 2	Numerics changed	Rejected because it crossed an unplanned quality boundary.

Negative results are part of the optimization map: they prevent rediscovery and show that smaller tensors, more overlap, more fusion, or a different encoder are not automatically faster at this shape.

D7: implemented, attempted, but deliberately unclaimed

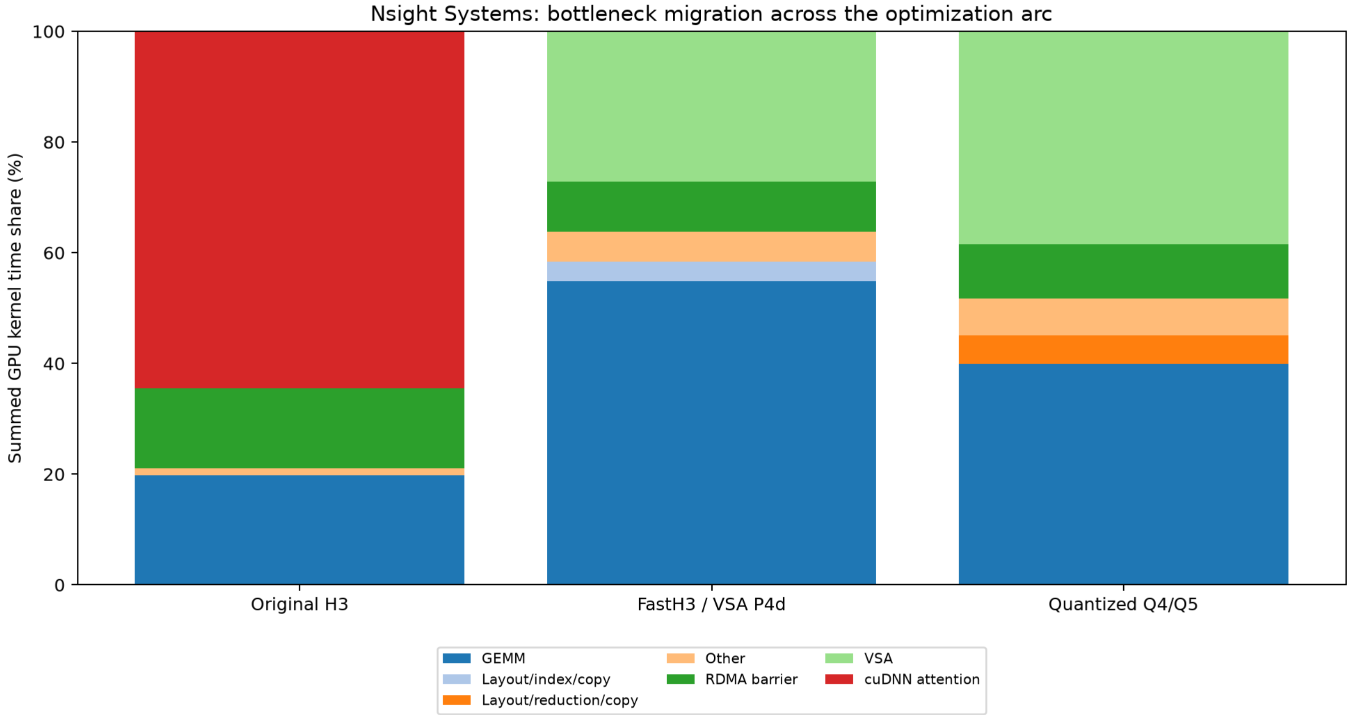
AAC pre-encoding and decode-focused NVTX ranges were implemented and tested. A formal A/B and final decoder-focused Nsight capture were attempted. All GPUs were briefly idle, then an external SGLang workload occupied GPUs 4-7. The runner was stopped before model load, so there is no D7 E2E number and no final decode Nsight trace.

The final accepted endpoint therefore remains D6 at 14.561 s. This is conservative: implementation status is not converted into benchmark evidence.

Nsight Systems methodology

Nsight Systems answers timeline questions: which subsystem owns the step, whether kernels and transfers overlap, which rank reaches a barrier last, and whether decode/audio/encode are serialized. Three valid DiT-era traces are available: original H3, FastH3/VSA P4d, and the quantized Q4/Q5-era stack.

Category GPU time is summed over ranks and streams and may exceed wall time. Barrier arrival-skew sum is already contained inside the step wall and must not be subtracted again.



Composition changes from cuDNN-attention-dominated to a near-even GEMM/VSA split.

Nsight Systems trace — Original H3

Metric	Value
Profiled step scope	11618.329 ms
Median active rank span	11597.613 ms
Barrier arrival-skew sum	1985.389 ms (17.09% of scope)
Source	results/vllm-omni-h3-sm120-sp8-step03-nsys-20260905T025828Z/h3-analysis.json

Kernel category	Summed GPU-time share
cuDNN attention	64.534%
GEMM	19.822%
RDMA barrier	14.363%
Other	1.281%

Interpretation: dense attention dominates; rank arrival imbalance is material. This trace justifies prioritizing the attention graph and communication path.

Nsight Systems trace — FastH3 / VSA P4d

Metric	Value
Profiled step scope	5225.109 ms
Median active rank span	5197.833 ms
Barrier arrival-skew sum	568.697 ms (10.88% of scope)
Source	results/vllm-omni-fasth3-vsa-p4d-sm120-sp8-step03-nsys-20260906T042700Z/h3-analysis.json

Kernel category	Summed GPU-time share
GEMM	54.849%
VSA	27.137%
RDMA barrier	9.023%
Layout/index/copy	3.485%
Other	5.506%

Interpretation: four-step distillation and VSA move the bottleneck toward GEMM. Sparse attention remains substantial, while layout/index/copy costs become visible enough to optimize.

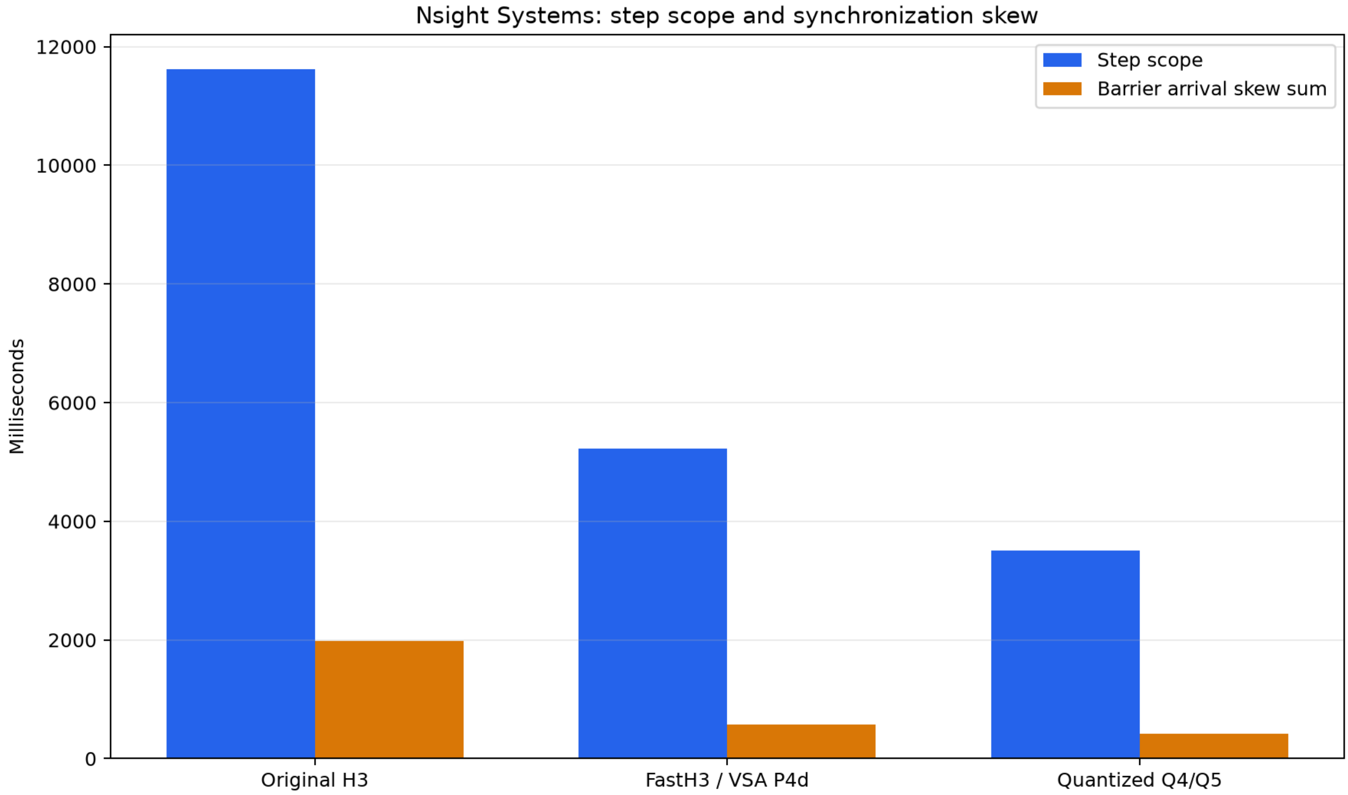
Nsight Systems trace — Quantized Q4/Q5

Metric	Value
Profiled step scope	3503.224 ms
Median active rank span	3485.235 ms
Barrier arrival-skew sum	423.197 ms (12.08% of scope)
Source	results/vllm-omni-fasth3-vsa-qkv-e4m3-mlp-out-qkv-fp8-o-bundle-sm120-sp8-step3-nsys-20260907T000000Z/h3-analy

Kernel category	Summed GPU-time share
GEMM	39.863%
VSA	38.473%
RDMA barrier	9.759%
Layout/reduction/copy	5.150%
Other	6.755%

Interpretation: FP8 lowers GEMM share until VSA is nearly co-dominant. Further progress requires shape-specific sparse scheduling, not a generic GEMM-only campaign.

Nsight Systems: scope and synchronization skew



The step shrinks sharply; the synchronization fraction remains material even as its absolute duration falls.

The original trace's dominant late rank was GPU 7, accounting for 38.35% of the recorded barrier-arrival skew. That diagnostic is only visible in a multi-rank timeline; per-kernel timing cannot identify it.

What genuinely requires Nsight Systems

Question	Why ordinary timers are insufficient
Are video and audio decode concurrent?	Only a cross-rank timeline proves overlap and exposes a hidden serialization gap.
Which rank gates a collective?	A mean kernel time hides last-arriving-rank skew.
Does D2H overlap encode?	Separate timers can double-count overlap; the timeline shows causality.
Is barrier residency network transfer?	Timeline context separates waiting/spinning from data movement.
Does profiler overhead explain a regression?	Compare profiler-on/off scopes and launch patterns.
Where is the remaining decoder critical path?	Nested NVTX ranges around TAEH3, Audio VAE, copies, encode and mux are required.

The last item remains instrumented but unmeasured for D6. The report does not substitute the illustrative rank diagram for a trace.

Decoder NVTX plan for the next clean profiling window

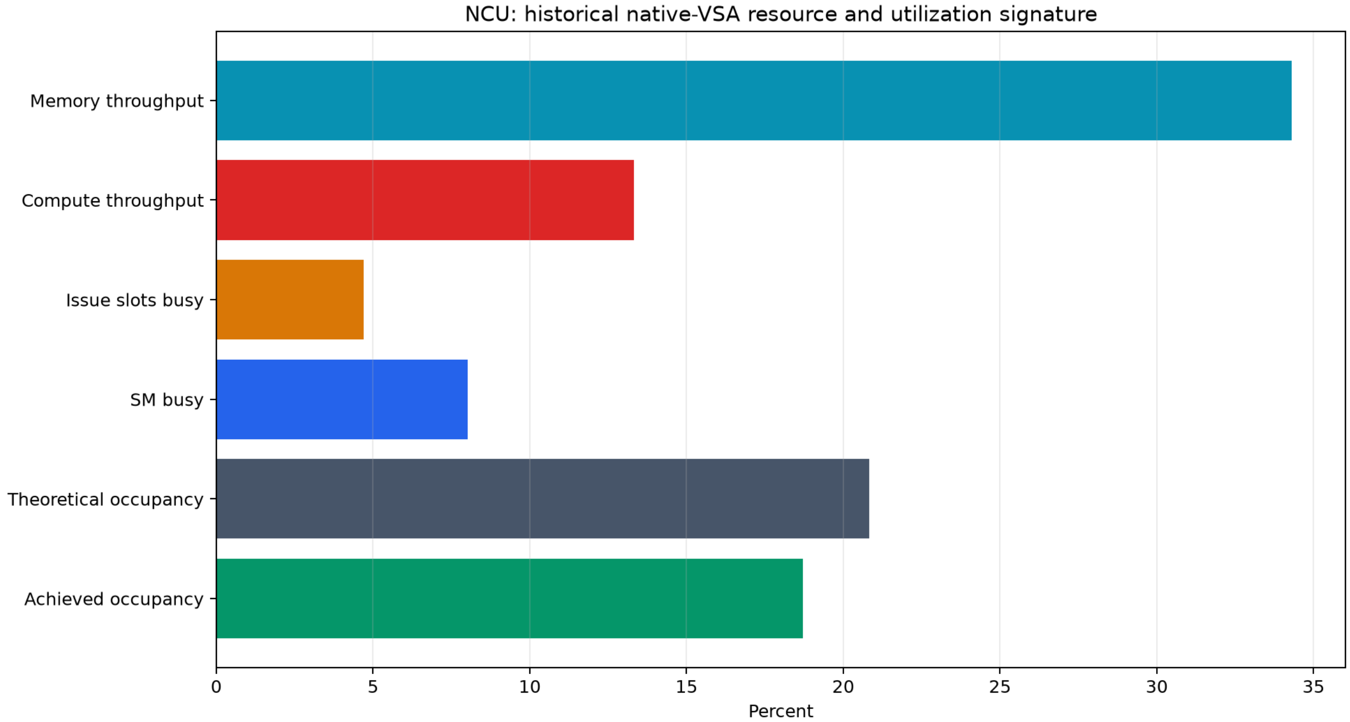
Range	Question answered
decode.video.taeh3	Exact TAEH3 GPU interval on rank 0
decode.audio.vae	Audio VAE interval on rank 1
decode.video.d2h	Pinned-copy duration and overlap
encode.video.pyav	CPU video encode interval
encode.audio.aac	AAC encoding placement
mux.mp4	Final serialization and filesystem completion
request.post	True endpoint wall boundary

Capture conditions: reserve all eight GPUs, warm once, record one formal request with CUDA/NVTX/OS runtime, export .sqlite, and run the nested-range analyzer. Do not accept a trace if another process enters the device set.

NCU methodology

Nsight Compute answers kernel-mechanism questions: occupancy limits, issue efficiency, cache behavior, shared-memory pressure, wavefronts, and instruction-level stalls. It does not establish request E2E or cross-rank overlap.

The historical native-VSA report and the final exact-general FlashInfer matrix have different scopes. They are reported separately and are never merged into one artificial speedup claim.



Historical native-VSA resource signature; useful for mechanism, not the final-kernel latency.

NCU: historical native-VSA bottleneck

Metric	Value
duration ms	93.570
memory throughput pct	34.310
dram throughput pct	0.810
l1 pct	39.250
l2 pct	5.180
compute pct	13.310
issue slots busy pct	4.690
sm busy pct	8.010
l2 hit pct	97.340
registers per thread	162
dynamic shared kb	49.150
theoretical occupancy pct	20.830
achieved occupancy pct	18.710
branch efficiency pct	97.510
excess shared wavefronts pct	79.000
warning	Historical native VSA experimental kernel; not the final exact-general FlashInfer kernel.

Registers (162/thread) and dynamic shared memory (49.15 KiB) each cap blocks per SM at two, limiting theoretical occupancy to 20.83%. Achieved occupancy is 18.71%. Low issue-slot and SM busy values, combined with 79% excessive shared wavefronts, point to scheduling/data-movement inefficiency rather than raw DRAM bandwidth.

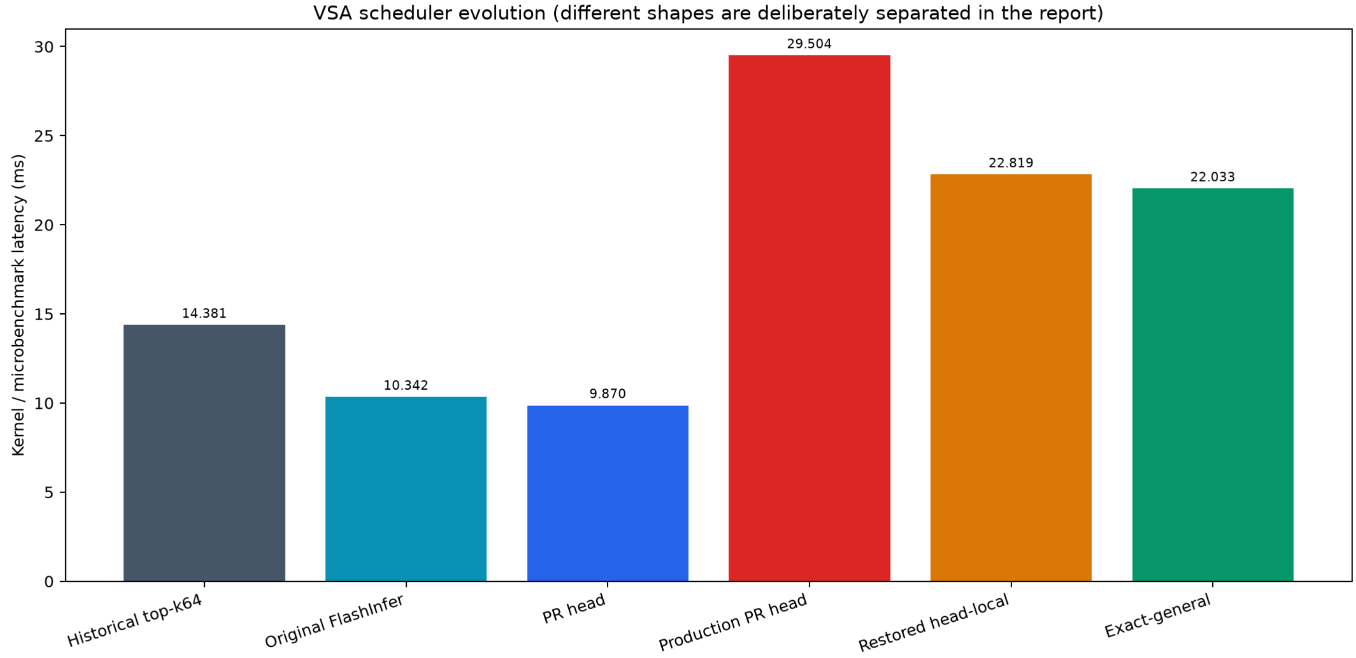
NCU: exact FlashInfer shape matrix

Field	Value
scope	single-GPU synthetic-QKV exact FlashInfer SM120 VSA fine-kernel shape matrix
status	passed
cases	3
geomean p50 ms	11.724132
max cv	0.0171457
hashes	all reference hashes matched

The three cases cover square 10-second geometry, the production-like 720p 15-second landscape geometry, and an SP4 head geometry. All reference hashes matched and the maximum coefficient of variation was 0.01715.

This matrix supports exactness and stability of the candidate fine-kernel path; it does not replace an E2E benchmark.

VSA scheduler evolution



Shape changes are shown, but only matched-shape pairs are used for direct percent claims.

For the production shape, restoring head-local scheduling improved 29.504 ms to 22.819 ms (~22.7%). Exact-general measured 22.032896 ms. In the final same-GPU paired microbenchmark, parent was 23.525841 ms and candidate 23.316159 ms (~0.89%), with identical output hashes.

What genuinely requires Nsight Compute

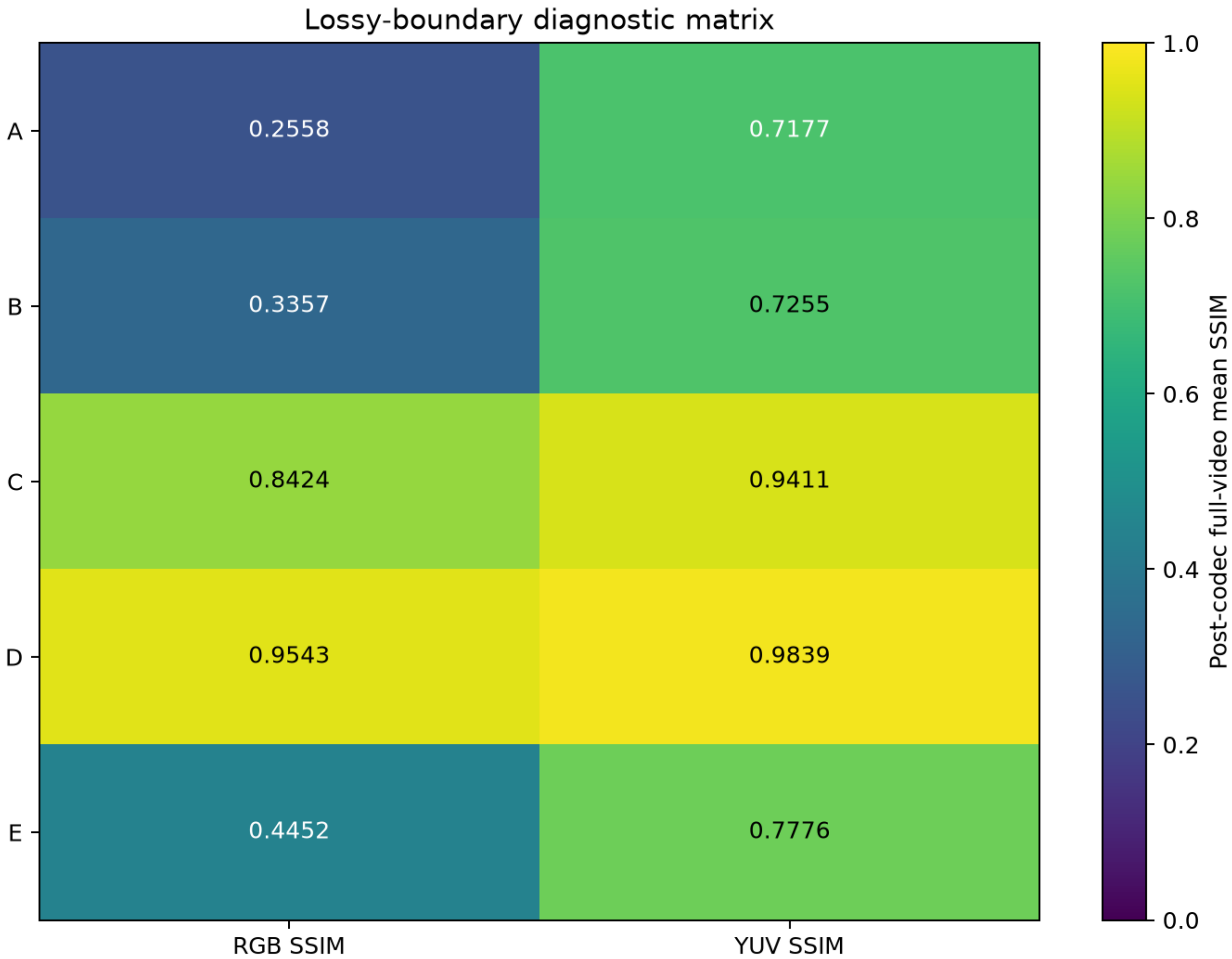
Question	Required NCU evidence
Why does KV32 regress?	Warp occupancy, branch divergence, sectors, waves and scheduler stalls.
Is a fusion register-bound?	Registers/thread, occupancy and spill metrics.
Is shared-memory tiling pathological?	Shared wavefronts, bank conflicts, dynamic shared bytes and block limits.
Does a new top-k shape underfill SMs?	Grid size, active warps, eligible warps and achieved occupancy.
Is L2 reuse hiding DRAM cost?	L2 hit rate, sectors and DRAM throughput together.
Did FP8 move the bottleneck?	Tensor-pipe utilization, instruction mix and memory traffic on matched shapes.

NCU should be reserved for a small, shape-locked kernel matrix because replay overhead makes it unsuitable for full E2E timing.

Quality audit method

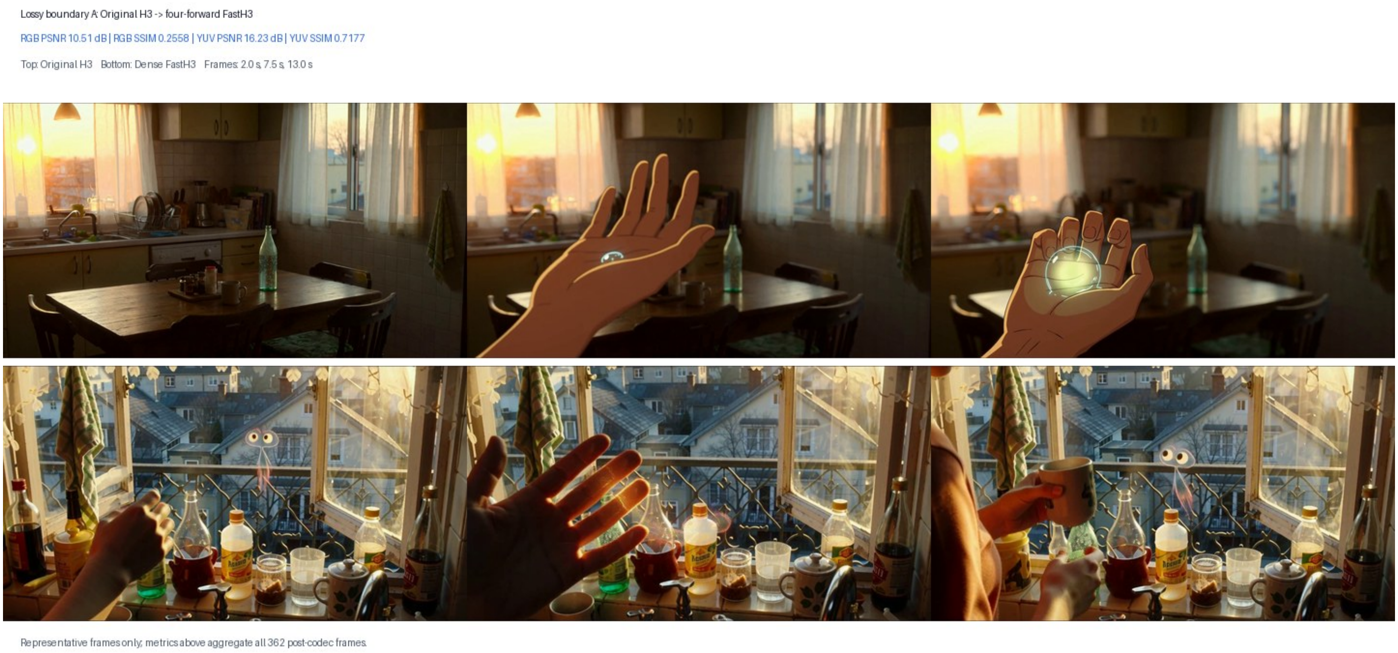
Each declared lossy boundary is represented by a real pair of persisted MP4s. Both files are validated as 362-frame, 1280x704, 24 fps H.264 with stereo 32 kHz AAC. FFmpeg deterministically converts both decoded streams to planar RGB and YUV444 before per-frame PSNR and SSIM. Audio is decoded to stereo interleaved float32 PCM.

PSNR/SSIM compare two stochastic-generation trajectories under one seed; they are diagnostic, not a human preference score and not an automated admission threshold. Low values after distillation or broad FP8 changes mean the trajectory diverged, not necessarily that the candidate is unusable.



All values are full-video post-codec means across 362 frames.

Lossy boundary A: Original H3 -> four-forward FastH3



Top row is the reference; bottom row is the candidate. Columns are 2.0 s, 7.5 s and 13.0 s.

Metric	Full-video result
RGB global PSNR	10.514 dB
RGB mean SSIM	0.255811
YUV global PSNR	16.234 dB
YUV mean SSIM	0.717655
Decoded PCM	not identical; overlap SNR 0.69 dB
Evidence SHA-256	57395900540362167e05ef0635943ea5e8da2a1fdc0d4a06b455ae2f4837c704

Distillation changes the denoising trajectory from 49 forwards to four; large pixel differences are expected. The contact sheet is the primary human-auditable record for scene retention.

Lossy boundary B: BF16 compute -> MLP FP8 + QKV E4M3

Lossy boundary B: BF16 compute -> MLP FP8 + QKV E4M3

RGB PSNR 12.68 dB | RGB SSIM 0.3357 | YUV PSNR 18.56 dB | YUV SSIM 0.7255

Top: BF16 control   Bottom: FP8 / E4M3   Frames: 2.0 s, 7.5 s, 13.0 s

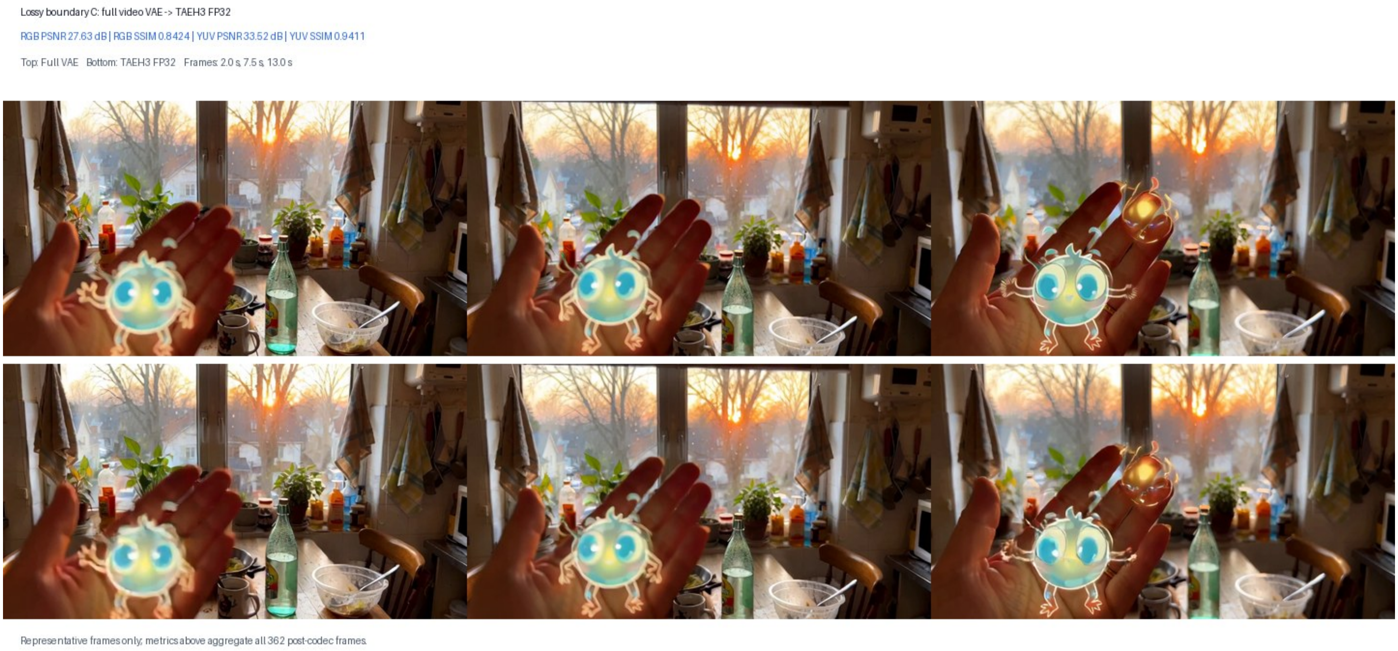
Representative frames only; metrics above aggregate all 362 post-codec frames.

Top row is the reference; bottom row is the candidate. Columns are 2.0 s, 7.5 s and 13.0 s.

Metric	Full-video result
RGB global PSNR	12.682 dB
RGB mean SSIM	0.335719
YUV global PSNR	18.560 dB
YUV mean SSIM	0.725505
Decoded PCM	not identical; overlap SNR 1.10 dB
Evidence SHA-256	fa4236a89d14024fa9458a8b8ac11a08293d4a023f7643328a11cdade78c4658

Online low precision perturbs the generated trajectory. This pair is not evidence of pixel identity; it quantifies divergence and exposes frames for inspection.

Lossy boundary C: full video VAE -> TAEH3 FP32



Top row is the reference; bottom row is the candidate. Columns are 2.0 s, 7.5 s and 13.0 s.

Metric	Full-video result
RGB global PSNR	27.634 dB
RGB mean SSIM	0.842449
YUV global PSNR	33.516 dB
YUV mean SSIM	0.941100
Decoded PCM	bitwise identical
Evidence SHA-256	896e781ca7300dc961c2cb3ab6f9d185bb8a659ed88b1826c0f15aa3662f768d

The video decoder alone changes; decoded audio PCM is bitwise identical, isolating the audiovisual boundary.

Lossy boundary D: TAEH3 FP32 -> TAEH3 FP16

Lossy boundary D: TAEH3 FP32 -> TAEH3 FP16

RGB PSNR 39.24 dB | RGB SSIM 0.9543 | YUV PSNR 45.22 dB | YUV SSIM 0.9839

Top: TAEH3 FP32 Bottom: TAEH3 FP16 Frames: 2.0 s, 7.5 s, 13.0 s

Representative frames only; metrics above aggregate all 362 post-codec frames.

Top row is the reference; bottom row is the candidate. Columns are 2.0 s, 7.5 s and 13.0 s.

Metric	Full-video result
RGB global PSNR	39.236 dB
RGB mean SSIM	0.954258
YUV global PSNR	45.223 dB
YUV mean SSIM	0.983943
Decoded PCM	bitwise identical
Evidence SHA-256	4bbd445af9d7a0ddcc8375d82f6a045c6fef2c7a6e173b6a5a541c058c87b249

This matched decoder-precision pair is the closest quality comparison: YUV mean SSIM 0.983943 and identical decoded audio PCM.

Lossy boundary E: partial FP8 -> all-main FP8

Lossy boundary E: partial FP8 -> all-main FP8

RGB PSNR 14.92 dB | RGB SSIM 0.4452 | YUV PSNR 20.83 dB | YUV SSIM 0.7776

Top: Partial FP8 Bottom: All-main FP8 Frames: 2.0 s, 7.5 s, 13.0 s

Representative frames only; metrics above aggregate all 362 post-codec frames.

Top row is the reference; bottom row is the candidate. Columns are 2.0 s, 7.5 s and 13.0 s.

Metric	Full-video result
RGB global PSNR	14.916 dB
RGB mean SSIM	0.445237
YUV global PSNR	20.826 dB
YUV mean SSIM	0.777641
Decoded PCM	not identical; overlap SNR -1.75 dB
Evidence SHA-256	e2b95e27085fce92fbe438301f8a12680d38ee3853e0becee599e387b9ef158b

Online low precision perturbs the generated trajectory. This pair is not evidence of pixel identity; it quantifies divergence and exposes frames for inspection.

Quality metric ledger

Boundary	RGB PSNR	RGB SSIM	YUV PSNR	YUV SSIM	Audio
A	10.51	0.2558	16.23	0.7177	SNR 0.69 dB
B	12.68	0.3357	18.56	0.7255	SNR 1.10 dB
C	27.63	0.8424	33.52	0.9411	identical
D	39.24	0.9543	45.22	0.9839	identical
E	14.92	0.4452	20.83	0.7776	SNR -1.75 dB

Additional historical comparisons: matched QKV-E4M3 output measured approximately SSIM 0.5658 / PSNR 16.41 dB; an older MLP-FP8 vs BF16 comparison measured video SSIM ~0.5808 / PSNR ~17.53 dB and corrected full-scale audio PSNR ~31.62-32.02 dB. They are retained as historical diagnostics, not thresholds.

Exactness and artifact identity ledger

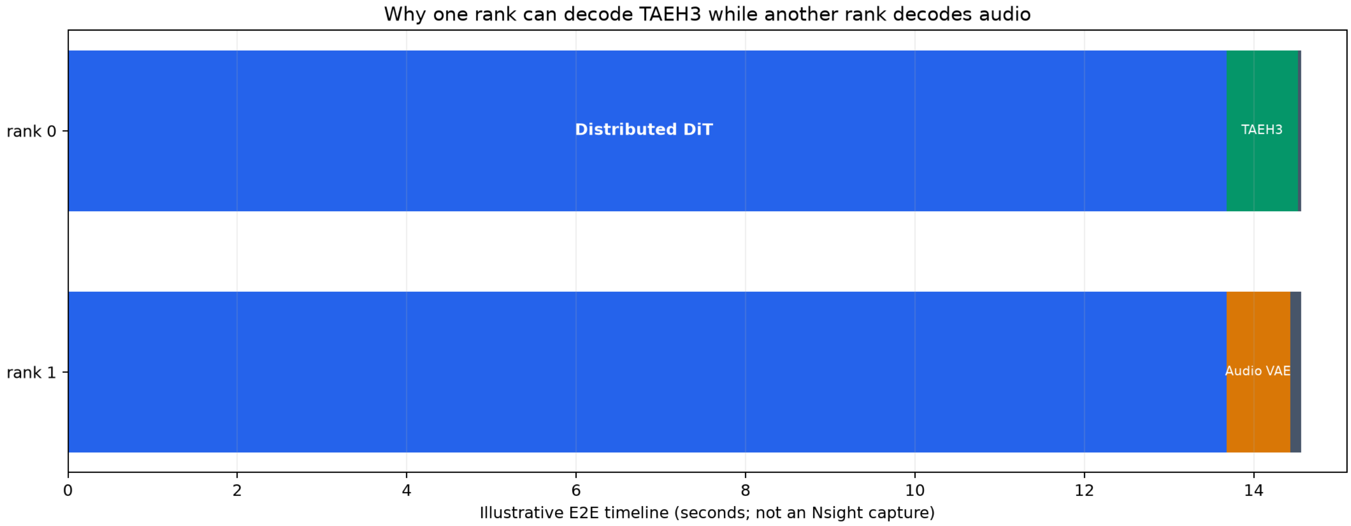
Boundary	Identity statement
Exact-general VSA fine kernel	All exact-matrix reference hashes matched; padding poison unchanged; return_lse=false buffer unchanged.
TAEH3 FP32 -> FP16	Not identical; dedicated 362-frame quality audit supplied.
D3 -> D4 -> D5 -> D6 systems changes	Selected pipeline semantics unchanged; final D6 media SHA-256 dbdbc19811ac656678648cc49555563f3fc31c2a
Audio for full VAE -> TAEH3 FP32	Decoded stereo-f32 PCM bitwise identical.
Audio for TAEH3 FP32 -> FP16	Decoded stereo-f32 PCM bitwise identical.
Prompt	Locked UTF-8 SHA-256 613e0a1beccf83566e3531852c52c2e11daffa3b255222cd67726cdf73c1a563.

Exactness is scoped to the chosen parent pipeline. It does not imply equality to original 49-forward H3 after earlier lossy boundaries.

Why TAEH3 needs only one rank

The original full video VAE is patch/tile parallel across eight ranks and contains collective communication. It needs every participating rank because each rank owns a spatial/temporal patch and collective exchanges reconstruct the full result.

TAEH3 is a different decoder topology. After the distributed DiT finishes, each rank has access to the required latent for rank-local temporal decoding. TAEH3 contains no patch-parallel collectives in this path, so rank 0 alone can decode the video. Rank 1 can simultaneously run Audio VAE on a different GPU. Remaining ranks do not need to join either decoder.



Conceptual overlap, derived from stage measurements; explicitly not a substitute for the pending final Nsight capture.

Cross-rank decoder critical path

D1 reduced full-VAE decode from about 6.484 s to 1.554 s. D2 reduced TAEH3 decode to about 1.050 s. D4 measured about 0.847 s and D5 about 0.835 s after steady-state/system tuning. D6 then moved Audio VAE to rank 1 so its independent terminal work overlapped rank-0 TAEH3.

Stage	E2E	DiT	Video decode	Critical-path change
Q5 full VAE	21.649	14.994	6.484	Collective full VAE
D1 TAEH3 FP32	16.738	15.010	1.554	Rank-local video decoder
D2 TAEH3 FP16	16.289	15.068	1.050	Lower decoder precision
D4 persistent	15.165	14.149	0.847	Steady-state resources
D5 OMP=28	14.671	13.677	0.835	Host orchestration
D6 cross-rank audio	14.561	not separately profiled	not separately profiled	Video/audio overlap

Remaining lossless optimization space

The remaining honest lossless headroom is probably measured in tens to low hundreds of milliseconds, not another order of magnitude. The final result is already below real time, and exact-general VSA plus the retained dataflow changes are already present.

Candidate	Expected scale	Evidence needed	Risk
Confirm/optimize terminal mux tail	10-100 ms	Final decode Nsight NVTX capture	Low
AAC pre-encode overlap (D7)	Unknown; likely small	Clean three-run A/B + timeline	Low semantics; measurement pending
Reduce rank barrier skew	Potentially 10-100+ ms	Per-rank Nsight arrival analysis	Medium topology sensitivity
Shape-specific exact VSA scheduling	Sub-percent to few percent kerne	Matched NCU + paired microbench + E2E	Medium maintenance
Pinned-buffer lifetime/allocator cleanup	Low tens of ms	Allocator trace and repeated E2E	Low
CPU affinity/NUMA placement	Low tens of ms	OS runtime trace + repeated E2E	Node-specific
CUDA graph expansion	Unknown	Launch-gap timeline + exactness tests	Medium memory/shape rigidity

Do not count current exact-general VSA, persistent RDMA, OMP=28, or cross-rank audio a second time: all are already in D6.

Optimization decision tree

Observation	Tool	Action rule
A subsystem consumes seconds of wall time	Nsight Systems	Change architecture/overlap before micro-tuning.
One kernel dominates and occupancy is low	NCU	Tune resource use and scheduling on the exact production shape.
A proposed approximation changes the trajectory	Media audit	Persist matched MP4s, screenshots and full-video metrics.
A delta is below run variance	Repeated E2E	Treat as noise or structural cleanup, not a speedup claim.
A run is contaminated	Process/GPU provenance	Reject E2E; salvage only independently valid scoped metrics.
An implementation lacks a run	None	Label implemented-only.

Validation and tests

Suite	Result
Combined decoder/NVTX/analyzer tests	126 passed
FastVideo contract tests	100 passed
Formatting/static checks	ruff, mypy and pre-commit passed in the recorded environment
shellcheck	Unavailable; no pass claim
Media validation	362 frames, 1280x704, 24 fps, H.264 + stereo 32 kHz AAC
Final repetitions	14.576 / 14.555 / 14.551 s
Final media SHA-256	dbdbc19811ac656678648cc49555563f3fc31c2a572956dd24de48aa9f8e624f

Tests establish software and artifact contracts; they do not turn a lossy model change into a lossless one.

Reproduction outline

The full model workspace is intentionally not copied into this report repository. The report preserves command scope, artifact paths, hashes and compact profiler summaries. Reproduction requires the corresponding vLLM-Omni/FastVideo/FlashInfer worktrees and model weights.

```
1. Reserve all eight SM120 GPUs and verify no foreign compute process. 2. Use node-default/unlocked clocks and the locked request/seed/prompt hash. 3. Warm once; run three profiler-off POST requests. 4. Probe every returned MP4. 5. Capture one separate Nsight Systems request with NVTX ranges. 6. Run NCU only on the production-shape sparse kernel matrix. 7. Re-run post-codec quality comparison for every lossy boundary.

Report generator: python scripts/minimax_h3_pro5000_cumulative_report/generate_report.py
```

Evidence inventory

Artifact class	Checked-in evidence
Cumulative measurements	report_data.json and this PDF
Lossy video comparisons	evidence/quality/*.json (full 362-frame arrays and summaries)
Screenshots	screenshots/*.jpg at 2.0, 7.5 and 13.0 s
Nsight Systems	evidence/nsys/summary.json plus source artifact paths
Nsight Compute	evidence/ncu/summary.json plus exact-matrix scope
Figures	figures/*.png
Landing page	index.html
Integrity	SHA256SUMS.txt

Raw .nsys-rep, .sqlite, .ncu-rep and source MP4s are not duplicated because they are large. Their derived summaries and quality comparison artifacts are sufficient to audit the claims made here, while source paths identify the originals.

Profiler source ledger

Original H3

```
results/vllm-omni-h3-sm120-sp8-step03-nsys-20260905T025828Z/h3-analysis.json
```

FastH3 / VSA P4d

```
results/vllm-omni-fasth3-vsa-p4d-sm120-sp8-step03-nsys-20260906T042700Z/h3-analysis.json
```

Quantized Q4/Q5

```
results/vllm-omni-fasth3-vsa-qkv-e4m3-mlp-out-qkv-fp8-o-bundle-sm120-sp8-step3-nsys-20260907T000000Z/h3
-analysis.json

NCU historical import: results/vsa-native-v1-real.ncu-rep

NCU exact matrix: results/profile/h3-vsa-exact-ncu-cold-candidate-forward-20260907/results.json

Analyzer: vllm-omni/tools/minimax_h3/analyze_h3_nsys_step.py

Decode profile wrapper: tools/minimax_h3/nsys_profile_h3_taeh3_cross_rank_decode.sh
```

Limitations

Limitation	Consequence
Single node/GPU class	Results do not automatically transfer to B300, H100, or another topology.
One locked prompt and seed	Latency is tightly controlled; quality/generalization requires a prompt suite.
Profiler traces stop before final decoder stack	D6 overlap is supported by E2E behavior and architecture, but not yet by a final Nsight timeline.
PSNR/SSIM only	They expose pixel trajectory differences but do not replace human preference or modern perceptual metrics.
Mixed bundles exist	Some E2E gains cannot be assigned to individual kernel edits.
Dynamic clocks	Node-default policy reflects deployment behavior but increases variance relative to locked clocks.

Final conclusion

The strict target is met: a 15.083-second audiovisual artifact is generated in a 14.561-second formal mean, with a 0.439-second margin.

The cumulative speedup is 44.903x and the latency reduction is 97.773%. The result is credible because the report keeps three categories separate: architectural approximations that require quality evidence, exact systems optimizations that preserve the selected pipeline, and unmeasured/rejected work that contributes no claimed seconds.

The next responsible action is not to stack speculative optimizations. It is to reserve a clean GPU window and capture the final decode-focused Nsight timeline, then test only the small lossless tails that the trace reveals. Until that evidence exists, D6 remains the final result.

END OF REPORT